



GEA Mechanical Equipment

Liquids to Value

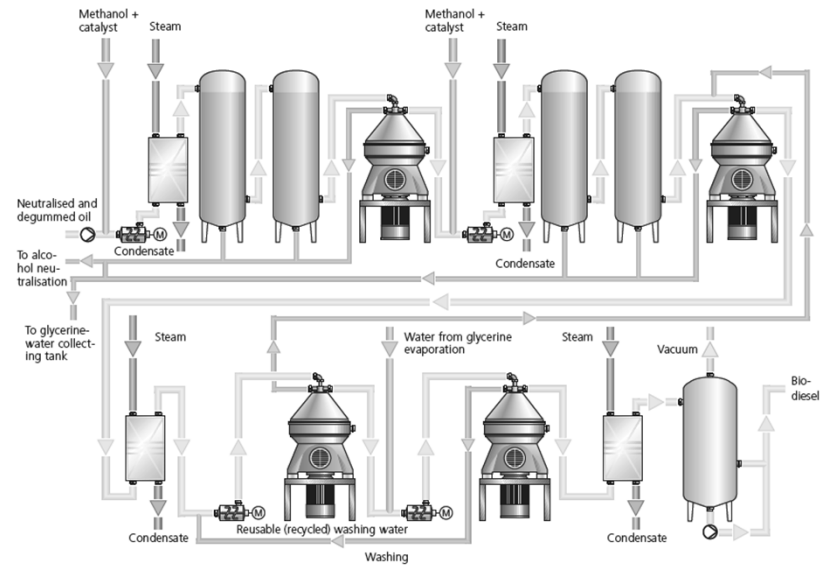
GEA Mechanical Equipment / GEA Westfalia Separator



Training Bio Diesel
Wilmar Nabati Indonesia
RSE 300 – 01 – 573
8th July 2011

Mechanical Equipment

Bio-diesel the fuel of the future



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Mechanical Equipment

General Information



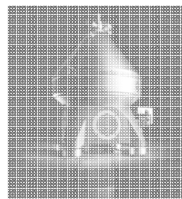
The machine described in this training is a high-speed centrifugal separator either with self-cleaning disc bowl or a solid bowl wall.

“Separation” means the separation of liquid mixtures which consist of two liquids, with simultaneous removal of solid contained in the liquids.

“Clarification” is the removal of solid from the liquid

Prerequisite for treatment technology (separation) is that the components of the product;

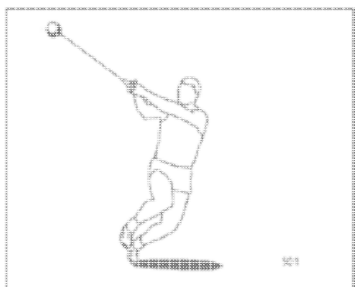
- Can be separated mechanically,
- Have different densities and,
- Do not emulsify



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Mechanical Equipment

General Information



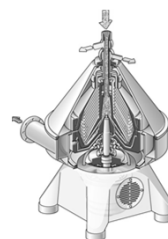
Under the influence of the centrifugal force, separation of the liquid mixture and /or ejection of the solid particles take place most rapidly.
The high centrifugal force is produced by very high bowl speed.

On one hand, high bowl speeds signify high efficiency, while on the other hand, they signify high material stressing of the bowl itself.

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Mechanical Equipment

General Information



8,000 g

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Mechanical Equipment

General Information



Important parameters for separation

- ⇒ **Physical force**
- ⇒ **Density difference between components**
- ⇒ **Particle size**
- ⇒ **Viscosity of liquids**
- ⇒ **Temperature**
- ⇒ **Time**

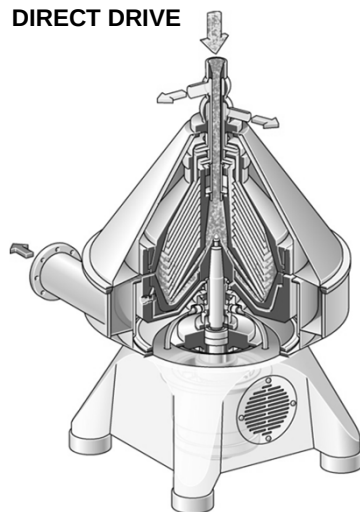
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Mechanical Equipment

Feature



RSE 300-01-573 DIRECT DRIVE



- **no dynamic clutch**
- **no gear drive**
- **no belt drive**
- **high efficiency** motor
- operation by **frequency converter**
- integrated **lubrication unit** for spindle bearing
- **motor access** from the top after dismantling of the bowl
- reduced **number of parts**
- **wewatch** remote monitoring for predictive maintenance system (option)

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Animation



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Separator in explosion-hazarded zones

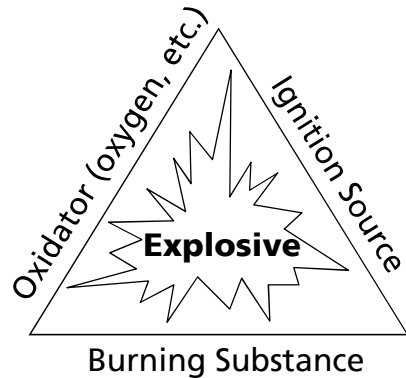
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Mechanical Equipment

Explosion possible



3 factors coming together to make an explosion possible



Oxidator
air atmosphere

Burning Substance
explosive gas
dust

Ignition Source
equipment

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Potential of ignition



A potential of ignition is in all types of centrifuges.

hot surface



- gas friction of the fast rotating bowl
- un-balance of the bowl
- damaged bearings or gaskets

mechanically induced sparks

- un-balance of the bowl
- heavily damaged bearings
- no proper adjusted or any broken parts

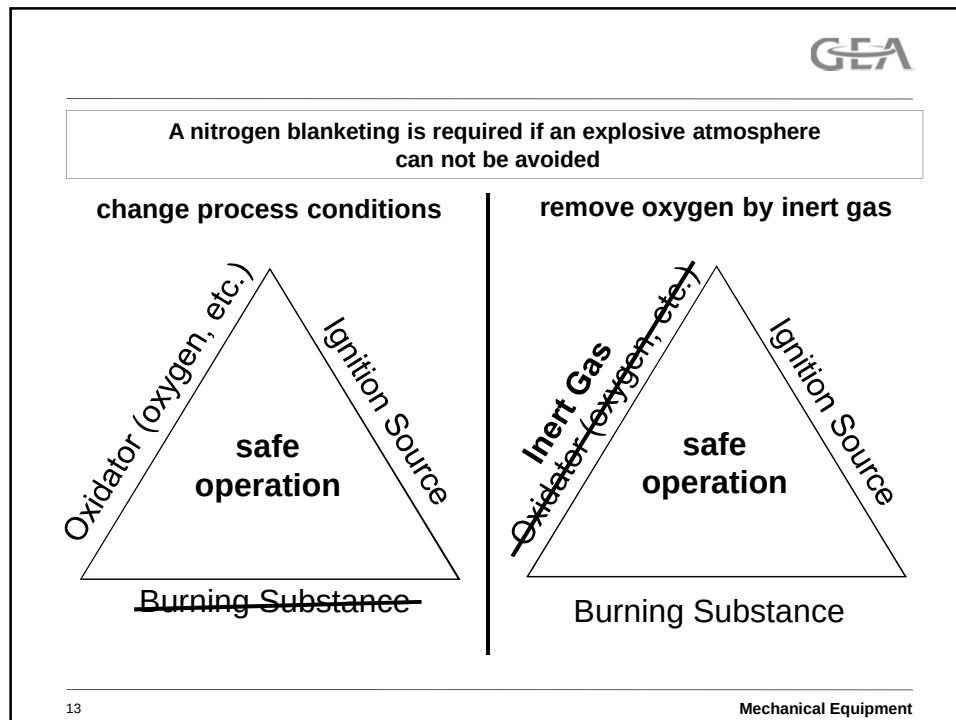
electrostatic loading



- high flow rates in small gaps
- spraying mist (aerosol)
- rubbing gaskets

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Gas Exchange calculation

Inerting the separator
 The hood and possibly drive chamber must be inerted before feeding easily flammable product to the separator. It is essential to pay attention to the following points;

- o Determine the highest admissible oxygen level.
- o The oxygen must be reduced to a level at which ignition is not possible.
- o The natural O₂ concentration in the air is 21%. After n-times gas exchange with inert gas the O₂ concentration falls as follow;

Gas exchange n = 1	produces a O ₂ – concentration :	10.50 %
Gas exchange n = 2	produces a O ₂ – concentration :	5.20 %
Gas exchange n = 3	produces a O ₂ – concentration :	2.60 %
Gas exchange n = 4	produces a O ₂ – concentration :	1.31 %
Gas exchange n = 5	produces a O ₂ – concentration :	0.65 %

Calculating the purging time

$$t = \frac{n \times V \times 60}{Q_{\max}} \quad [\text{min}]$$

t = Purging time [min]
 n = Number of gas exchanges
 V = Volume of the complete chamber with all connected chamber
 Q_{max} = throughput capacity during purging [l/h]

Nitrogen quality
 The purity of nitrogen must be at least 99 %
 The admissible dew point should not exceed 60°C to 80°C .
 Nitrogen which is too dry, shortens the service life of the separator seal.

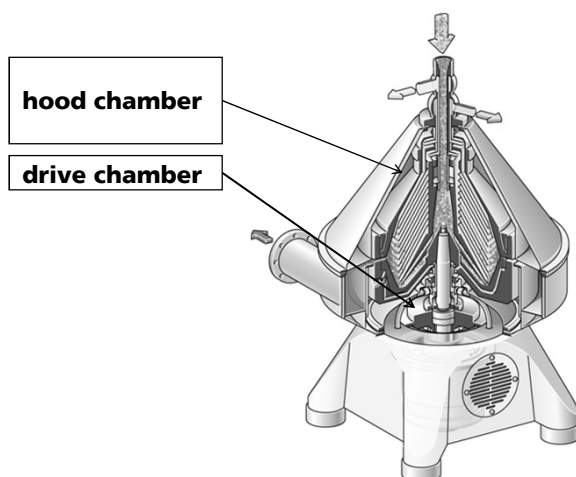
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Position of blanketing chamber



The following chambers are blanketed with inert gas:



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Pre-condition



Pre-flushing.

Before starting the drive motor, the chamber to be inerted must be purged with inert gas. The purging gas capacity to the hood and drive chamber are monitored by the flow switch. And the inerted pressure in the sealing gas range, drive and hood chamber are monitored by the pressure switch.

If, however the N₂ flow,

- to the drive range < 5000 l/h and / or
- to product range 9 – 211 < 12000 l/h and / or
- to product range 9 – 212 < 11500 l/h

And / or the inerted pressure in the

- drive range 9-262 <15 mbar and / or
- product range 9-222 < 6 mbar

A "difference hardware – software" alarm is triggered after the pre-flushing time elapse

The drive motor must not be started until the chamber have been purged completely without interruption.

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Mechanical Equipment

Pre-condition



Post-flushing.

During run-up of the drive motor, the chamber to be inerted will remain purged with inert gas. A pressure gauge monitors the hood, drive and sealing chamber for minimum pressure.

- If the inerted pressure in the;
 - sealing gas range [9-241] <15 mbar
 - and / or
 - product range [9-221] <10 mbar
 ➤ an alarm/warning is triggered after some time
- If the inerted pressure in the;
 - drive range [9-262] <15 mbar
 - and / or
 - product range [9-222] < 6 mbar
 ➤ the separator will be switched off after some time

Note, on start-up after an interruption in production.

If the blanketing pressure has not dropped below the minimum pressure at any time since the machine was last operated, the drive motor starts without purging / pre-flushing time.

Purging / post flushing is however carried out during run-up.

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Mechanical Equipment

Pre-condition



Monitoring during machine operation

Perfect operation is monitored as follows;

- A flow meter monitors the inert gas stream towards the hood and drive chamber.
 - When the preset limit value for maximum flow is exceeded
 - product range 9-211 >3000 l/h
 - and / or
 - drive range 9-252 > 3000 l/h
 ➤ an alarm/warning is triggered after some time
- A pressure gauge monitors the hood, drive and sealing chamber for minimum pressure.
 - If the inerted pressure in the;
 - sealing gas range [9-241] <15 mbar
 - and / or
 - product range [9-221] <10 mbar
 ➤ an alarm/warning is triggered after some time
 - If the inerted pressure in the;
 - drive range [9-262] <15 mbar
 - and / or
 - product range [9-222] < 6 mbar
 ➤ the separator will be switched off after some time
 - the separator will be switched off.

The cause of the pressure drop **MUST** be found out and remedied.

Otherwise:



Leakage arising from the under pressure developing inside the separator might cause air to penetrate into the blanketed areas and form an explosive gas mixture. Together with an ignition source, this can lead to an implosion or explosion.

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N2 purge system



What are monitored during the purge time ?

1. N2 purge capacity --> monitored by the flow switch
2. Inerted pressure --> monitored by the pressure switch

What are monitored during the machine operation ?

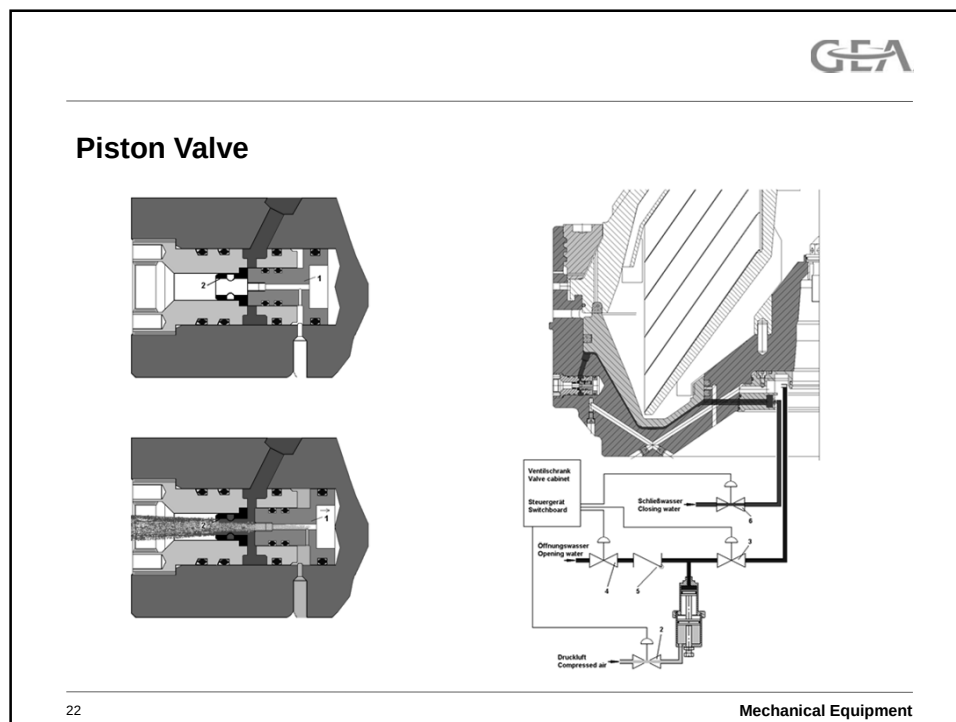
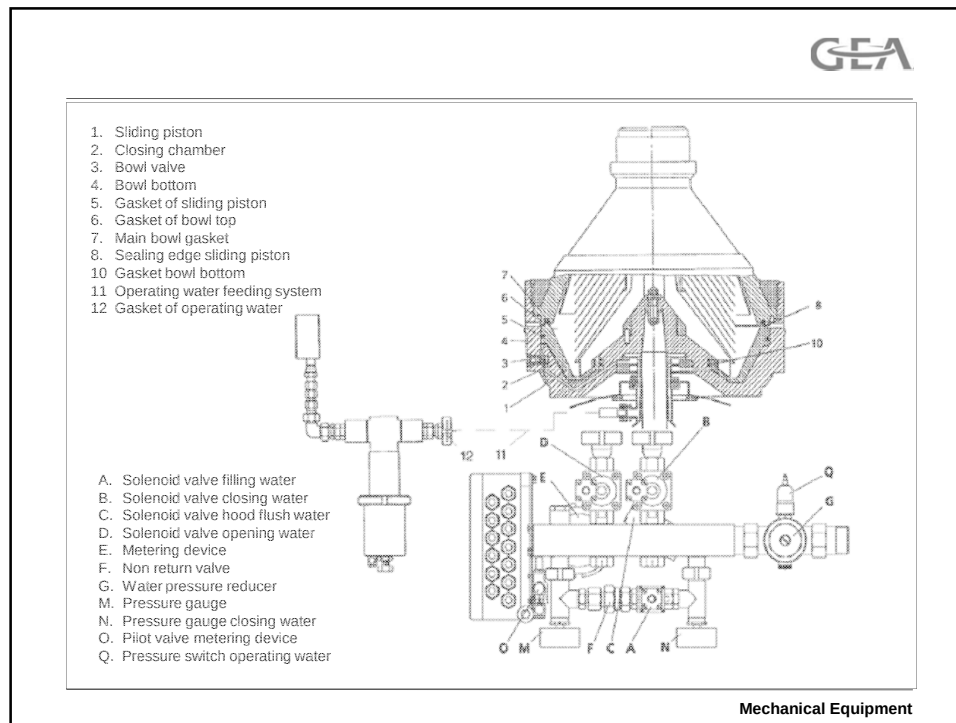
1. N2 purge capacity, monitored by the flow switch
2. Inerted pressure, monitored by the pressure switch
3. The bowl speed
4. Motor temperature
5. Bearing temperature
6. Machine vibration
7. Operating water pressure
8. Light phase discharge pressure

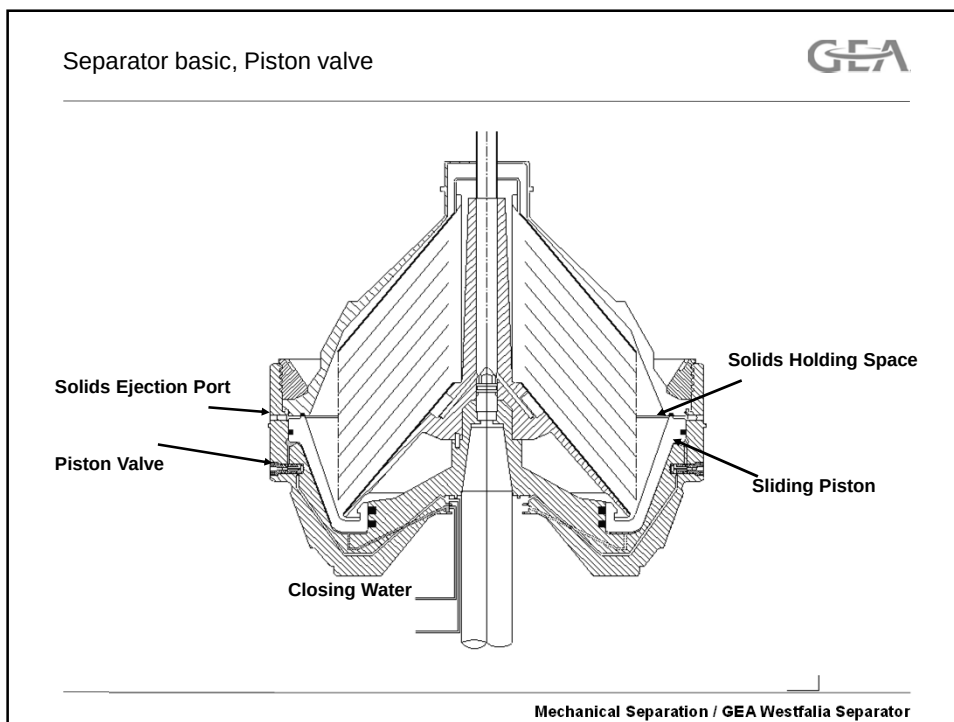
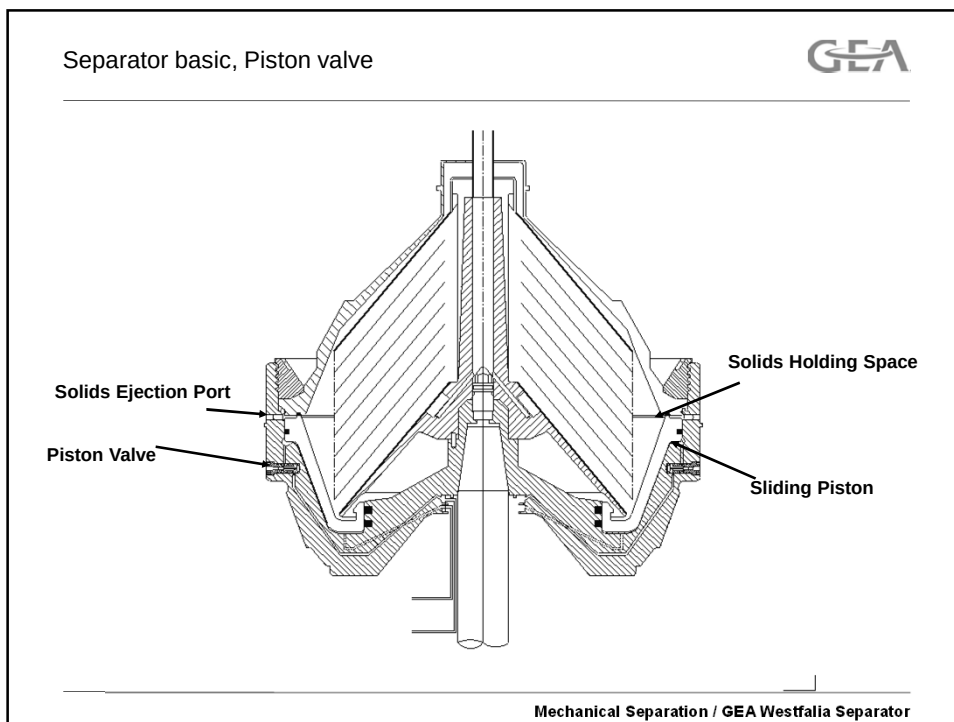
Mechanical Equipment

Bowl - animation

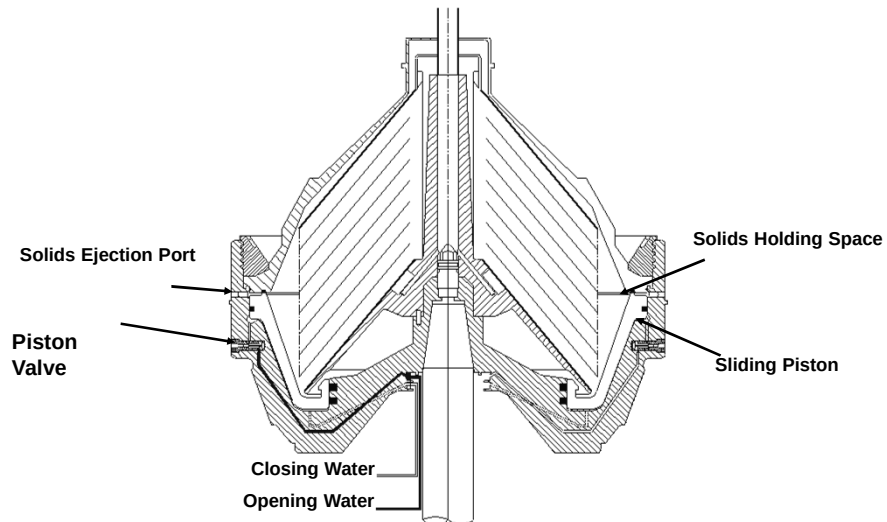


Mechanical Equipment



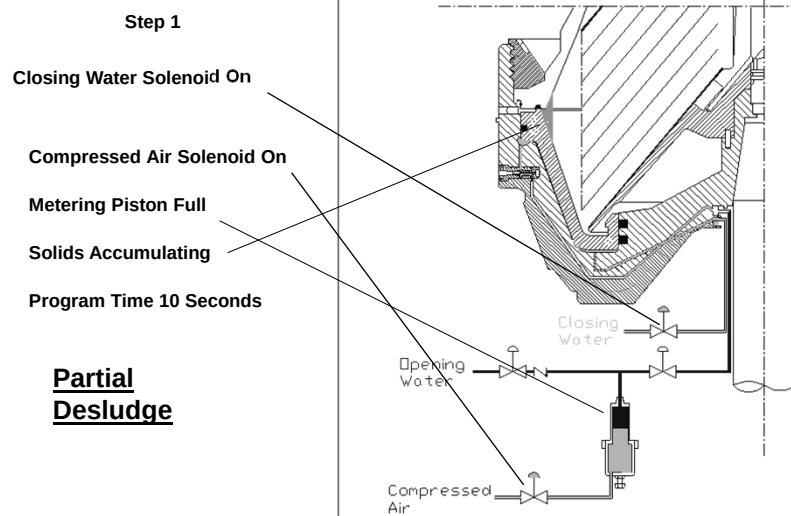


Separator basic, Piston valve



Mechanical Separation / GEA Westfalia Separator

Separator basic, Piston valve



Mechanical Separation / GEA Westfalia Separator

Separator basic, Piston valve

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Step 2

Closing Water Solenoid On

Compressed Air Solenoid On

Opening Water Solenoid On

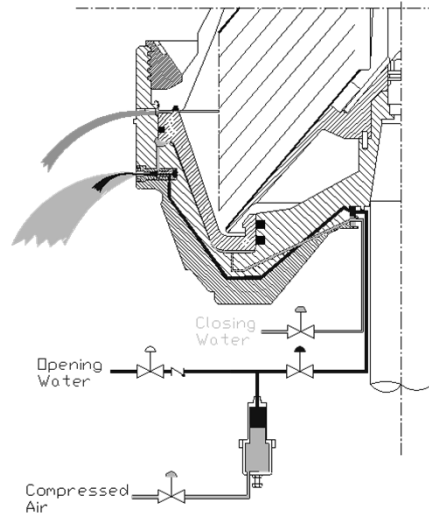
Metering Piston Activated

Solids & Water Eject

Program Time 2 Seconds

Partial
Desludge

Sliding Piston Open Approx. 0.5mm



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Separator basic, Piston valve

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Step 3

Closing Water Solenoid On

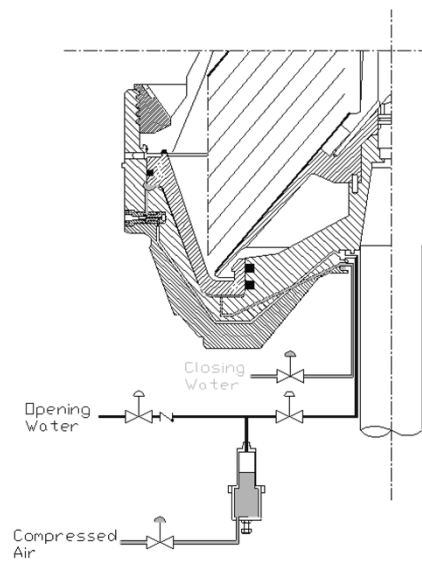
Compressed Air Solenoid On

Metering Piston Empty

Bowl Closed

Program Time 2 Seconds

Partial
Desludge



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Separator basic, Piston valve



Step 4

Closing Water Solenoid On

Compressed Air Solenoid On

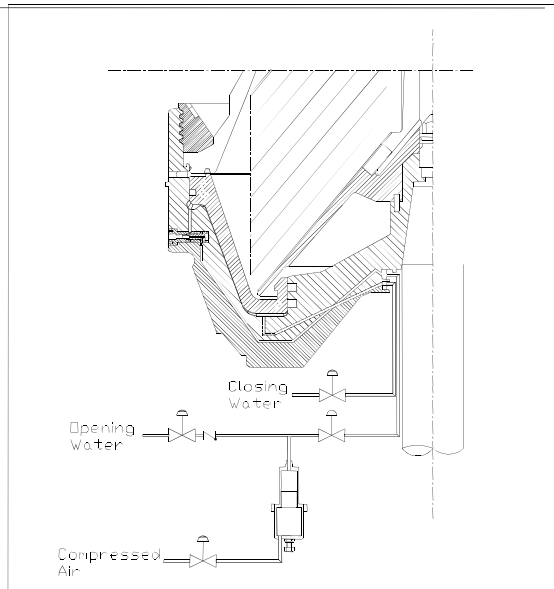
Metering Piston Water Solenoid On

Metering Piston Empty

Bowl Closed

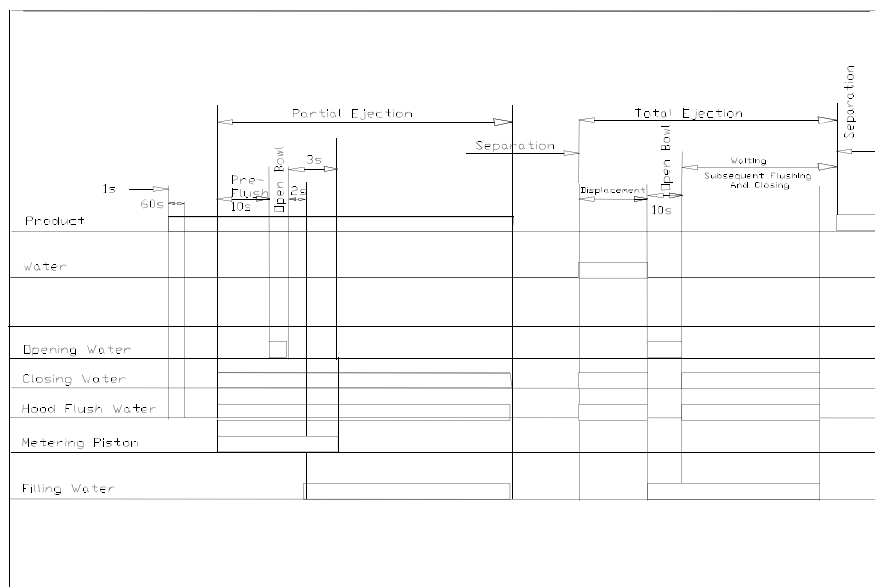
Program Time 1 Second

Partial
Desludge



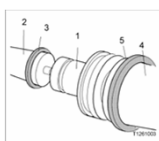
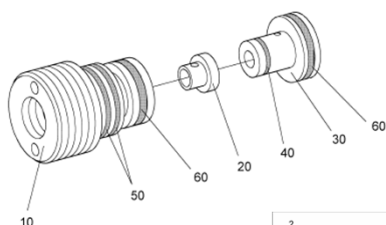
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Separator basic, Piston valve

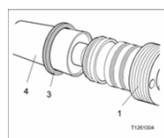


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Piston valve



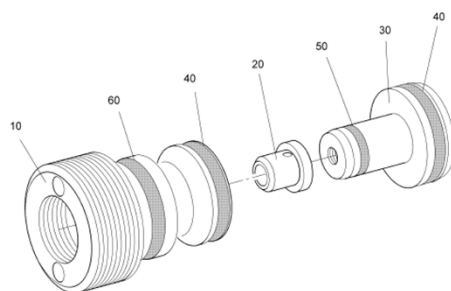
- Stretch gaskets 3 and 5 over expanding mandrel 2 and 4 and fit in piston 1.



- Stretch the gasket 3 over the expanding mandrel 4 and fit in piston valve 1.

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Piston valve

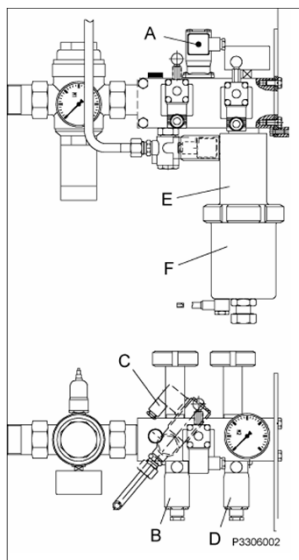


Westfalia Separator AG

Datum:
03.03.03.Zeichnungs-Nr.:
3395-4756-002Blatt 1
1 Blätter

Mechanical Equipment

Valve block



- A. Solenoid valve for filling water
- B. Solenoid valve for closing water
- C. Solenoid valve for hood flush water
- D. Solenoid valve for opening water
- E. Metering unit
- F. Air chamber

Operating water connection: 1 1/4 in
 Operating-water pressure: $P_{min.} = 2.0$ bar
 (with open closing water valve) $P_{max.} = 2.5$ bar
 Operating water capacity: = min. 4000 l/h
 Air pressure for metering unit: $P_{min.} = 4.0$ bar
 $P_{max.} = 6.0$ bar

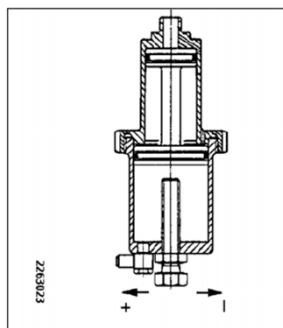
Mechanical Equipment

Metering piston



Metering piston

The amount of solid to be ejected can be adjusted externally during operation between 5 – 30 l is primarily determined by the opening water volume.
 The opening water volume is adjusted on the metering device.



Adjusting the amount to be ejected by means of the adjusting screw of the metering device.

Reduce ejection volume:

- Turn adjusting screw clockwise.

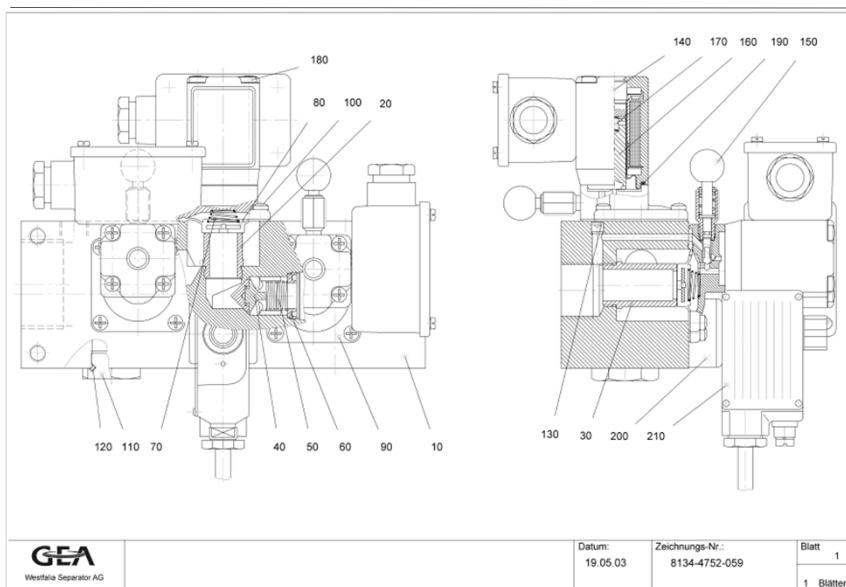
Increase ejection volume:

- Turn adjusting screw anti-clockwise

Mechanical Equipment

Solenoid valve

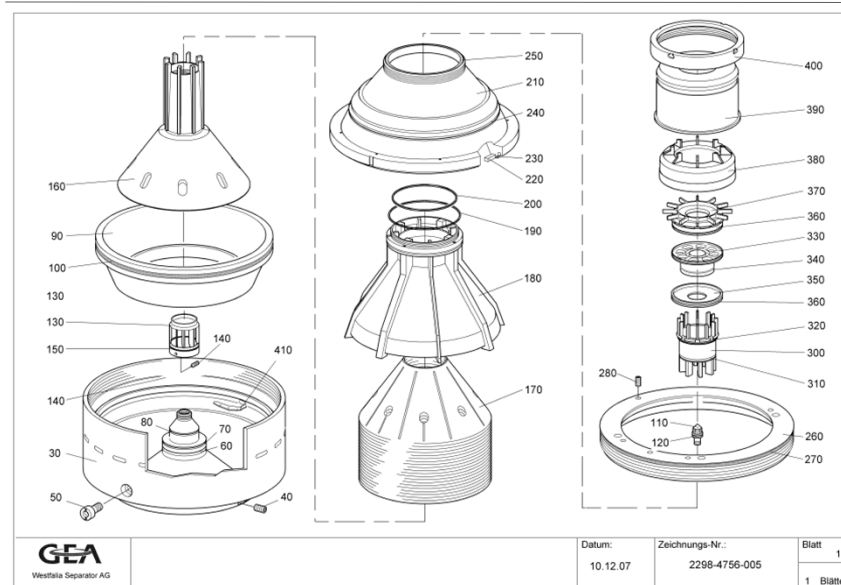
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Mechanical Equipment

Bowl complete

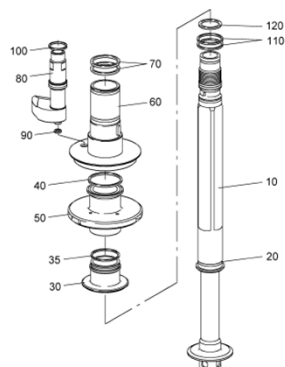
GEA



Mechanical Equipment

Centripetal pump

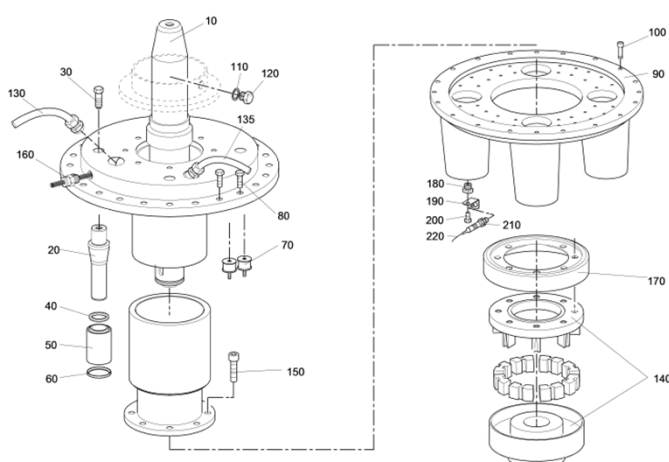
GEA

GEA
Westfalia Separator AGDatum:
09.08.2001Zeichnungs-Nr.:
2298-4752-002Blatt
1
1 Blätter

Mechanical Equipment

Drive assembling

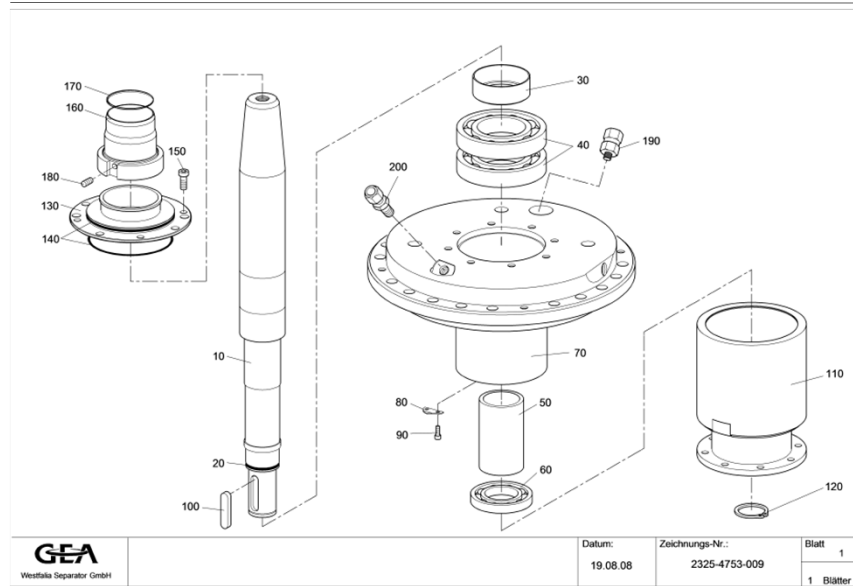
GEA

GEA
Westfalia Separator AGDatum:
28.08.08Zeichnungs-Nr.:
2325-4753-010Blatt
1
1 Blätter

Mechanical Equipment

Drive parts

GEA

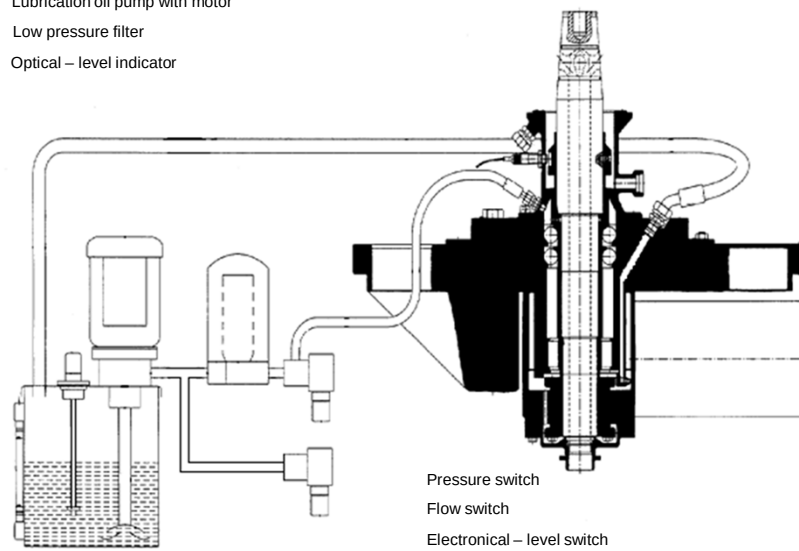


Mechanical Equipment

External lubrication unit

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Lubrication oil pump with motor
 Low pressure filter
 Optical – level indicator



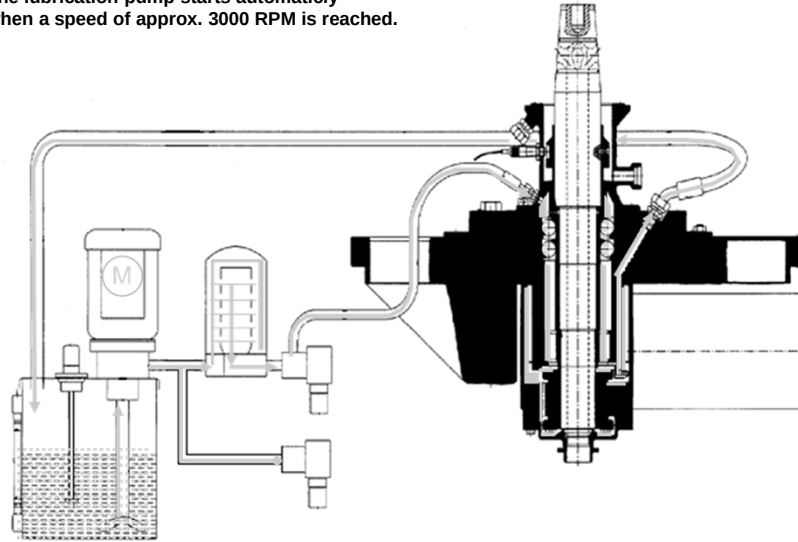
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Mechanical Equipment

External lubrication unit



The lubrication pump starts automatically when a speed of approx. 3000 RPM is reached.



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Mechanical Equipment

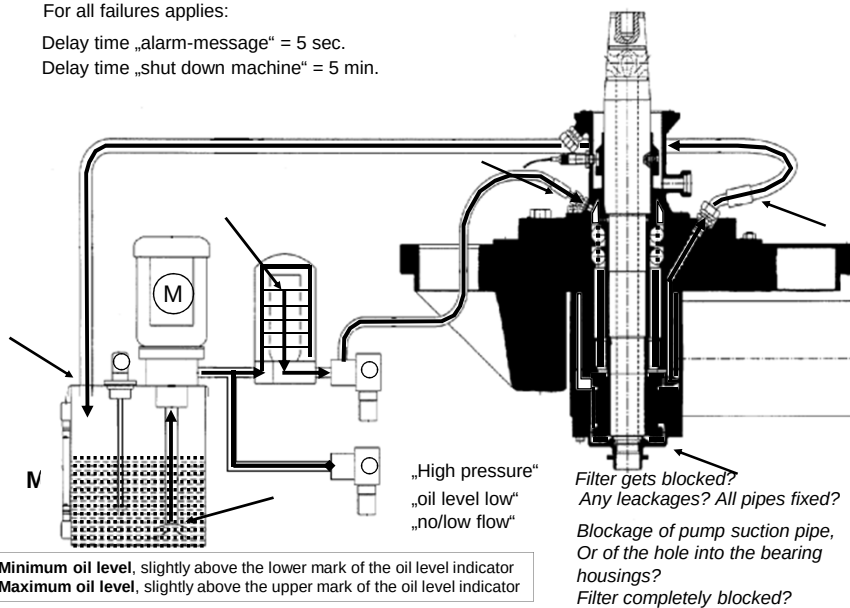
external lubrication unit



For all failures applies:

Delay time „alarm-message“ = 5 sec.

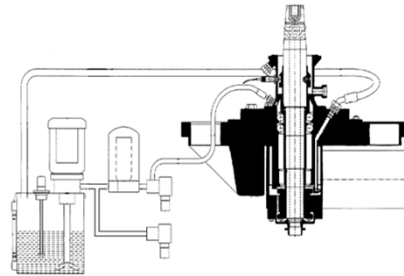
Delay time „shut down machine“ = 5 min.



External lubrication unit



What are the operators have to check:

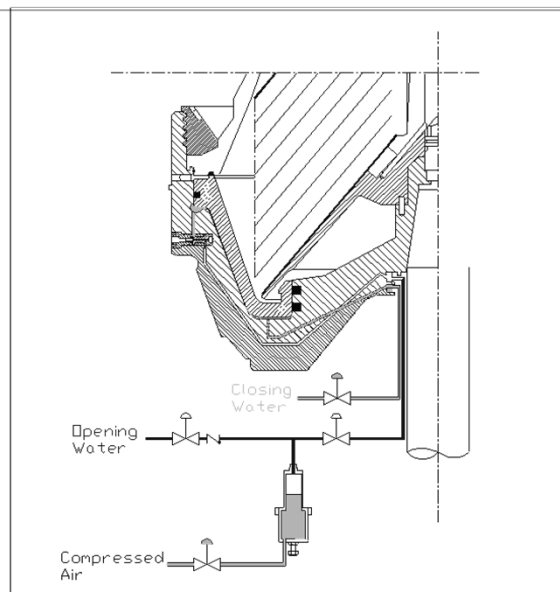


- Check the oil level and the circulation daily and top on oil if necessary!
(The oil must never drop under minimum level during operation.)
- Check from time to time whether the oil contains water!
- If the oil exhibits a milky colouring (emulsification), change the oil immediately!
(Try to trace the cause of the emulsification.)
- The oil needs to be replaced every 6000 operating hours, or at least once per year.

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Mechanical Equipment

Closing water pulse



Mechanical Separation / GEA Westfalia Separator



Timer no.	Designation	Description
T0 (720 s)	Separator start-up	This timer start when the separator is switched on, when this timer has run down and two total ejections (closing program) have been performed, the separator is ready for operation. The timer value adjusted must be a little higher than the start-up time of the separator
T01 (300 s)	Separation partial ejection	The timer start for the first time when the prod. Mode starts (pressing of the product function key or external signal) After expiry of the timer, either a partial ejection is started or a total ejection if timer T55; separation total ejection is expired
T02 (360 s)	Closing pulse pause	When the separator is ready for operation, closing water is fed into the bowl in the preset pulse/pause ratio to compensate for any loss due to leakage. The interval times stop during an ejection
T03 (1 s)	Closing pulse	
T04 (0 s)	Bowl closing before partial ejection	The closing water chamber of the bowl is topped up during this time
T26 (10 s)	Total ejection	Opening time of the bowl for a total ejection
T34 (10 s)	Pre-flushing hood	The flushing of the hood starts during a partial ejection
T37 (2 s)	Partial ejection	Bowl opens for partial ejection (10 s max.)

Mechanical Equipment



Timer no.	Designation	Description
T40 (10 s)	Post flushing hood	This timer start after a partial or total ejection and continues to actuate the hood flushing water
T42 (2 s)	Metering piston filling water	The filling water valve is actuated for the set time. After a fixed delay period of 1second the metering piston is pretension for 2 seconds
T44 (30 s)	Discharge pressure delay for product start	Upon activating the product program the PAL Discharge pressure fault message is delayed by this time. This enables bowl filling and the separating zone can appear
T45 (10 s)	Discharge pressure bridging	When partial or total ejection is complete, the PAL discharge pressure message is suppressed during this period
T55 (60 min)	Separation total ejection	After expiry of this timer and of timer T01; Separation partial ejection, a total ejection is carried out.
T56 (9999 min)	CIP overflow / Total ejection	During CIP; After this time an overflow followed by a total ejection is triggered
T57 (45 s)	Sep. start-up from min. speed	Only with speed monitoring pre-selection This timer start during start-up of the separator, when the minimum bowl speed is reached (see. Section of bowl speed limits and check this parameter) After this timer has run down, the closing program starts and the separator is ready for operation. The closing program starts after T0 or T57, whichever has run down first

Mechanical Equipment



Timer no.	Designation	Description
(5 s)	Operating water PAL	The program time delays the pressure controller signal in the operating water line. If the pressure falls below the set value beyond this time, a message is issued.
(5 s)	Control air PAL	The program time delays the pressure controller signal in the control air line. If the pressure falls below the set value beyond this time, a message is issued.
(5 s)	Analog inputs: • Discharge pressure PAL • Speed • Current	Dropping below the measuring range by <4mA is delayed by the set time. After this, a message is issued.
(5 s)	Feed back: • Drive • Valve	The feed back from the drives and valves (optional) are delayed by the set time when activating and deactivating these unit. After expiry of this time and no feedback is received, a corresponding message is issued.

 Mechanical Equipment


Timer no.	Designation	Description
(000 s)	Pre-flushing	This timer start when the separator is switched on, the purge valve activated, if no interruption either by flow alarm low nor pressure alarm low, the motor run up will carried out after this timer elapse.
(s)	Product range FAL 9-211	The program time delays the pressure controller signal in the control air line. If the pressure falls below the set value beyond this time, a message is issued.
(s)	Product range FAH 9-211	The program time delays the pressure controller signal in the control air line. If the pressure falls below the set value beyond this time, a message is issued.
(s)	Product range warning PAL 9-221	The program time delays the pressure controller signal in the control air line. If the pressure falls below the set value beyond this time, a message is issued.
(s)	Product range shut down 9-222	The program time delays the pressure controller signal in the control air line. If the pressure falls below the set value beyond this time, a message is issued.
(s)	Product range FAL 9-211	The program time delays the pressure controller signal in the control air line. If the pressure falls below the set value beyond this time, a message is issued.

 Mechanical Equipment

Fault message



Message	Explanation	Possible reason	Program action	Precondition	Reference
Emergency stop	Signal from emergency relay is active	- Normal condition after PLC initialization - Emergency stop operated - Switch defective	Emergency stop	can be happen in all condition	Wiring diagram 8551-6300-027 Sh. 5
Difference hardware-software	Difference timer pre-flushing (software and hardware – 1K30) > 60 sec	- Setting time of the software and hardware timer - check the N2 supply - check the inert pressure	Pre-flushing aborted	Pre-flushing	Wiring diagram 8551-6300-027 Sh. 28
Product range PAL 9_221 PIAL	N2 inert pressure in the product range < 10 mbar	- cable broken - sensor defective - line leakage	Info / Warning	separator ON	Wiring diagram 8551-6300-027 Sh. 26
Product range PAL 9_222 PISL	N2 inert pressure in the product range < 6 mbar	- cable broken - sensor defective - line leakage	Separator OFF	separator ON	Wiring diagram 8551-6300-027 Sh. 26
Product range FAH 9-211	N2 flow to product range > 3000l/h	- Check N2 - Cable defective - Incorrect cable to control cabinet	Info - Warning	Separator ON	Wiring diagram 8551-6300-027 Sh. 23

Mechanical Equipment

Fault message



Message	Explanation	Possible reason	Program action	Precondition	Reference
Product range FAL 9-211 FIAL / AH	N2 flow to product range < 12000l/h	- Check N2 supply - Cable defective - Incorrect cable to control cabinet	Pre-flushing abort	N2 pre-flushing	Wiring diagram 8551-6300-027 Sh. 23
Product range FAL 9-212 FISL	N2 flow to product range < 11500l/h	- Check N2 supply - Cable defective - Incorrect cable to control cabinet	Pre-flushing abort	N2 pre-flushing	Wiring diagram 8551-6300-027 Sh. 23
Sealing gas range PAL 9_241 PIAL	N2 inert pressure in the sealing gas range < 15 mbar	- Check N2 - Cable defective - Incorrect cable to control cabinet	Info - Warning	Pre-flushing or post flushing or separator ON or centrifuged inert	Wiring diagram 8551-6300-027 Sh. 22
Drive range FAL 9_252 FISL / AH	N2 flow to the gear box < 5000l/h	- Check N2 - Cable defective - Incorrect cable to control cabinet	Info - Warning	Pre-flushing or post flushing	Wiring diagram 8551-6300-027 Sh. 24
Drive range FAH 9_252 FISL / AH	N2 flow to the gear box > 3000l/h	- Check N2 - Cable defective - Incorrect cable to control cabinet	Info - Warning	Separator ON	Wiring diagram 8551-6300-027 Sh. 24
Drive range PAL 9_262 PISL	N2 inert pressure in the gear box < 15 mbar	- Check N2 - Cable defective - Incorrect cable to control cabinet	Separator OFF	Pre-flushing or post flushing or separator ON or centrifuged inert	Wiring diagram 8551-6300-027 Sh. 27

Mechanical Equipment

Fault message



Message	Explanation	Possible reason	Program action	Precondition	Reference
Vibro test not successful	Initial test of vibro sensor unit	Vibro sensor failure	Emergency stop	Bowl stand still	Wiring diagram 8550-7305-039 Sh. 27
Separator startup	The bowl nominal speed has not been reached in time	- Timer startup monitoring to short - FC incorrect setting - sensor defective	Separator OFF	- Separator ON - Timer T.00 (000s) is expired	Wiring diagram 8550-7305-039 Sh.13
Operating water pressure PAL	The operating water pressure has drop below the limit	- PRV blocked - Pressure switch defective	Program OFF	- Product ON - Timer T.44 (10s) is expired	Wiring diagram 8550-7305-039 Sh. 22
Discharge pressure PAL	Pressure discharge below the set value	- Pressure switch defective - Incorrect cable to control cabinet	Program OFF	- Product ON - Timer T.44 (10s) is expired	Wiring diagram 8550-7305-039 Sh. 22
FC feedback	No feedback signal from FC to PLC	- signal cable broken - FC fault - breaker trip	Separator OFF	Separator ON	Wiring diagram 8550-7305-039 Sh. 11
Analog input current	No signal to PLC	- control cable broken - FC fault	Separator OFF	Separator ON	8550-7305-039 Sh. 10 8550-0705-121 Sh.10

Mechanical Equipment

Fault message



Message	Explanation	Possible reason	Program action	Precondition	Reference
Motor TAH	The motor temp has been exceeded	- Sensor broken - The motor has been run over current for longer time	Emergency stop	Separator OFF	Wiring diagram 8550-0705-121 Sh. 13
Vibration step. 1	The vibration 3,5mm/s has been exceeded	- Bowl unbalance - Mechanical problem	Separator OFF	Total ejection (if this message persist for > 5 min an emergency stop follow)	Wiring diagram 8550-7305-039 Sh. 27
Vibration step.2	The vibration 8.0 mm/s has been exceeded	- Bowl unbalance - Mechanical problem	Program OFF	Emergency stop (without bowl ejection)	Wiring diagram 8550-7305-039 Sh. 27
Speed input	No pulse from initiator when separator is switched on	- Sensor defective - Cable defective - Speed relay defective	Separator OFF	Separator OFF	Wiring diagram 8550-7305-039 Sh. 13
Over speed	The max permissible bowl speed has been exceeded	- incorrect setting on speed relay - incorrect frequency	Separator OFF	Separator OFF	Wiring diagram 8550-7305-039 Sh. 14

Mechanical Equipment

Vibration Sensor



5.2. Switch Functions

- | | |
|---------------|--|
| SW1, SW2 | Measuring range settings |
| SW3, SW4, SW5 | Selects output read-out |
| SW6, SW7 | Latch or Auto-reset function of Alert and Danger |
| SW8 | Connects housing (chassis) of Vibration Guard to Power GND |

Mechanical Equipment



Thank you very much for your attention!



Liquids to Value

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